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Tau hyperphosphorylation impairs cooperative binding to microtubules and perturbs organelle trafficking in neurons

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eLife Assessment

In their **valuable** study, Beaudet, Berger and Hendricks provide a mechanistic link between disease-associated tau hyperphosphorylation, loss of cooperative tau envelope formation on microtubules, and dysregulation of axonal transport prior to aggregation. Using complementary in vitro reconstitution and human iPSC-derived neuronal assays with phosphodeficient and phosphomimetic tau constructs targeting 14 disease-relevant sites, the authors **convincingly** show that phosphorylation state alters tau organization on microtubules and differentially impacts kinesin- and lysosome-based transport. The evidence is **solid** and well aligned with the conclusions, yet the work could be further strengthened by incorporating additional controls and motor-specific assays to refine the mechanistic depth.

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Abstract

Tau, a neuronal microtubule-associated protein (MAP), organizes the axonal cytoskeleton and regulates intracellular transport. Tau hyperphosphorylation is linked to neurodegeneration in tauopathies including Alzheimer's disease. Tau binds microtubules cooperatively to form cohesive tau envelopes, which are thought to control access to the microtubule lattice and regulate the activity of motor proteins and other microtubule-associated proteins. However, how disease-related perturbations affect tau dynamics and its function as a selective barrier to intracellular transport remains unclear. Using tau phosphomutants in vitro and in live neurons, we show that tau hyperphosphorylation disrupts cooperative binding and dysregulates lysosome transport. Hyperphosphorylated tau does not form envelopes, distributes more uniformly along the axon, and dissociates faster from microtubules. Tau weakly inhibits KIF5C motility, but strongly inhibits KIF1A. Hyperphosphorylation reduces KIF5C inhibition but increases KIF1A inhibition by decreasing processivity and accelerating detachment. Consistent with these effects, hyperphosphorylated tau alters lysosome transport in neurons. While phospho-resistant tau inhibits processive lysosome motility, hyperphosphorylated tau weakens tau-mediated regulation of lysosome transport—mimicking tau knockout neurons that exhibit enhanced processivity. Altogether, these findings show that hyperphosphorylation disrupts tau envelopes and impairs lysosome trafficking, likely contributing to early defects in degradative pathways that drive neurodegeneration.

Introduction

Tauopathies are a class of neurodegenerative diseases characterized by the misregulation and pathological aggregation of tau, a microtubule-associated protein (MAP) highly expressed in neurons (Binder et al., 1985). Tau regulates the axonal cytoskeleton (Kanai et al., 1992; Chen et al., 1992; Panda et al., 2003; Rosenberg et al., 2008; Chung et al., 2015; Biswas and Kalil, 2018), and acts as a selective barrier on microtubules to direct intracellular transport (Vershinin et al., 2007; Dixit et al., 2008; Vershinin et al., 2008; McVicker et al., 2011; Hoeprich et al., 2014; Hoeprich et al., 2017; Chaudhary et al., 2018; Monroy et al., 2018; Tan et al., 2019; Siahaan et al., 2019; Monroy et al., 2020; Beaudet et al., 2024). Tau's function and microtubule affinity is modulated by alternative splicing (Kellogg et al., 2018; McVicker et al., 2014) and phosphorylation, both of which play key roles in normal development and pathology (Lindwall and Cole, 1984; Mandelkow et al., 1995; Trinczek et al., 1995; Hasegawa et al., 1998; Niewidok et al., 2016; Kanaan and Grabinski, 2021). In Alzheimer's disease and other tauopathies, tau becomes abnormally hyperphosphorylated, reducing its microtubule affinity and promoting aggregation (Schneider et al., 1999; Cho and Johnson, 2003). While tau hyperphosphorylation is a hallmark of disease, its impact on the microtubule cytoskeleton and axonal transport, and in turn neuronal proteostasis, remains unclear.

Tau exhibits both diffusive and cooperative interactions with microtubules. Cooperativity leads to cohesive envelope-like structures (Dixit et al., 2008; Hinrichs et al., 2012; McVicker et al., 2014; Siahaan et al., 2019; Tan et al., 2019; Siahaan et al., 2022). Tau envelopes act as selective, reversible barriers along microtubules, regulating the access of motor proteins and other MAPs to the microtubule. Envelope formation is directed by the intramolecular microtubule lattice spacing between α - β tubulin. Tau does not form envelopes on GMPCPP-polymerized microtubules, which are irreversibly expanded, mimicking the GTP-state, but readily forms envelopes on compacted GDP-lattices or reversibly expanded taxol-lattices (Tan et al., 2019). Most studies investigating tau dynamics have relied on *in vitro* reconstitution assays using purified systems and stabilized microtubules. However, it remains unclear how the nucleotide state distribution and dynamic properties of microtubules in cells influence tau binding. Axonal microtubules are thought to be highly stable (Black et al., 1984; Brady et al., 1984; Sahenk and Brady, 1987; Lim et al., 1989; Okabe and Hirokawa, 1990; Baas et al., 1991; Ahmad et al., 1993; Li and Black, 1996; Song et al., 2013), but the rate of tubulin turnover and precise nucleotide and lattice compaction landscape of microtubules throughout the axonal cytoskeleton is not well-characterized. Further, while tau cooperative binding is well-established in purified, *in vitro* systems, it was an open question if tau forms envelopes in neurons.

The extent of tau's influence on motor proteins depends on several factors including *i*) tau isoform and posttranslational modifications (PTMs), *ii*) the underlying microtubule lattice, and *iii*) the intrinsic properties of the motors themselves. While some motors, such as kinesin-1 and kinesin-3, are strongly inhibited by tau in an isoform- and lattice-dependent manner, others, like kinesin-2 and dynein, can more effectively navigate tau-decorated regions (Dixit et al., 2008; Vershinin et al., 2007; McVicker et al., 2011; Hoeprich et al., 2014; McVicker et al., 2014; Siahaan et al., 2019; Tan et al., 2019; Siahaan et al., 2022). These studies offer important mechanistic insights into how tau regulates individual motor proteins, but questions remain: how do tau's regulatory effects scale when multiple motors work in coordinated ensembles to transport organelles? Do additional behaviors of tau, like envelope formation, emerge *in vivo* that alter transport dynamics? What is the impact of tauopathy-related perturbations on motor protein function?

Using *in vitro* reconstitution and live-cell imaging in iPSC-derived neurons, we find that disease-associated hyperphosphorylation alters tau dynamics on microtubules and impairs intracellular transport. We used phosphomutants in which 14 disease-associated residues were mutated to glutamate (E14) to mimic hyperphosphorylation or alanine (AP) to prevent phosphorylation. Our results show that while WT and AP tau bind microtubules cooperatively, forming envelopes both *in vitro* and in live neurons, E14 tau interacts more diffusely, has altered axonal localization, and

fails to form envelopes. Hyperphosphorylation also affects tau-mediated kinesin regulation. WT and AP tau only mildly inhibit KIF5C motility but increase detachment within envelopes, whereas E14 tau weaken these effects. In contrast, KIF1A is strongly inhibited by either unphosphorylated or phosphorylated tau. Furthermore, in live tau-knockout neurons, lysosomes move more processively than in control neurons. Rescue assays showed that hyperphosphorylation phenocopies tau knockouts, weakening tau-mediated regulation of lysosome transport, resulting in increased processive motility. Combined, our results demonstrate how disease-related hyperphosphorylation weakens the formation of cooperatively-bound tau envelopes on microtubules and perturbs tau-mediated regulation of degradative organelles, which would be expected to cause defects in cargo maturation and protein homeostasis, contributing to neurodegeneration.

Results

Hyper-phosphomimetic tau exhibits diffusive microtubule interactions and reduced tau envelope formation in vitro

We examined how phosphorylation governs tau cooperative binding and envelope formation along microtubules. The phosphostate of tau is strongly dependent on the expression system. Bacterial tau lacks PTMs, while insect and mammalian systems add variable and inconsistent phosphorylation (Siahaan et al., 2024 [↗](#); Fan et al., 2025 [↗](#)), making it hard to compare results and understand how disease-related phosphorylation affects tau function. To address this, we used in vitro reconstitution assays with mammalian-expressed phosphomutants of tau that either mimic or block phosphorylation at 14 serine and threonine residues commonly phosphorylated in Alzheimer's disease (Fulga et al., 2007 [↗](#); Steinhilb et al., 2007 [↗](#); Hoover et al., 2010 [↗](#)). We compared WT tau with hyper-phosphomimetic tau (E14), in which these 14 residues were mutated to glutamate, and phosphorylation-resistant tau (AP), in which they were mutated to non-polar alanine (Fig 1A [↗](#)). This system allowed us to directly assess the impact of disease-associated hyperphosphorylation on tau's microtubule binding without the added complexity and variability introduced by PTMs in other expression systems.

We expressed GFP-tagged tau constructs in COS-7 cells and prepared cell lysates. Lysates were adjusted to ensure an equal tau concentration of 500 nM before they were added to flow chambers containing taxol-stabilized microtubules immobilized on glass coverslips (Fig 1B [↗](#)). Our initial observations, consistent with previous studies, showed that tau binds to microtubules in two distinct kinetic phases: cooperative binding that results in the formation of cohesive envelopes and diffusive binding along the microtubule (Hinrichs et al., 2012 [↗](#); McVicker et al., 2014 [↗](#); Stern et al., 2017 [↗](#); Tan et al., 2019 [↗](#); Siahaan et al., 2022 [↗](#); Cario et al. 2022 [↗](#); Siahaan et al., 2024 [↗](#); Fan et al., 2025 [↗](#)). Consistent with previous studies, we defined the two distinct populations of tau along microtubules as (1) tau envelopes, marked by regions of bright pixel intensity that form sharp boundaries with adjacent gap regions containing (2) diffusive tau, characterized by dim pixel intensity (Fig 1B [↗](#)). To quantify the effects of hyperphosphorylation on tau-microtubule interactions, we applied a non-biased approach using Gaussian Mixture Models (GMMs). Tau-GFP intensity profiles were generated by performing linescans along individual microtubules and analyzed with a GMM to identify distinct tau populations. A threshold based on the first peak of the GMM distribution plus two standard deviations (Fig 1C [↗](#)), was then used to segment regions of high intensity (envelopes) from regions of lower intensity (gaps) (Fig 1B [↗](#) and C [↗](#)). We found that AP tau formed more robust and frequent envelopes compared to WT tau, whereas E14 tau showed significantly reduced intensity and frequency of envelopes (Fig 1D [↗](#) and E [↗](#)).

To examine the effects of hyperphosphorylation on tau's cooperative binding and microtubule binding in the absence of microtubule lattice compaction, we repeated our experiments using GMPCPP-polymerized microtubules. Previous work demonstrated that tau's cooperative binding on taxol-stabilized microtubules is facilitated by its ability to compact the intramolecular spacing between tubulin dimers. Taxol reversibly expands the microtubule lattice, and tau can displace taxol from the microtubule, allowing it to induce lattice compaction. In contrast, GMPCPP-

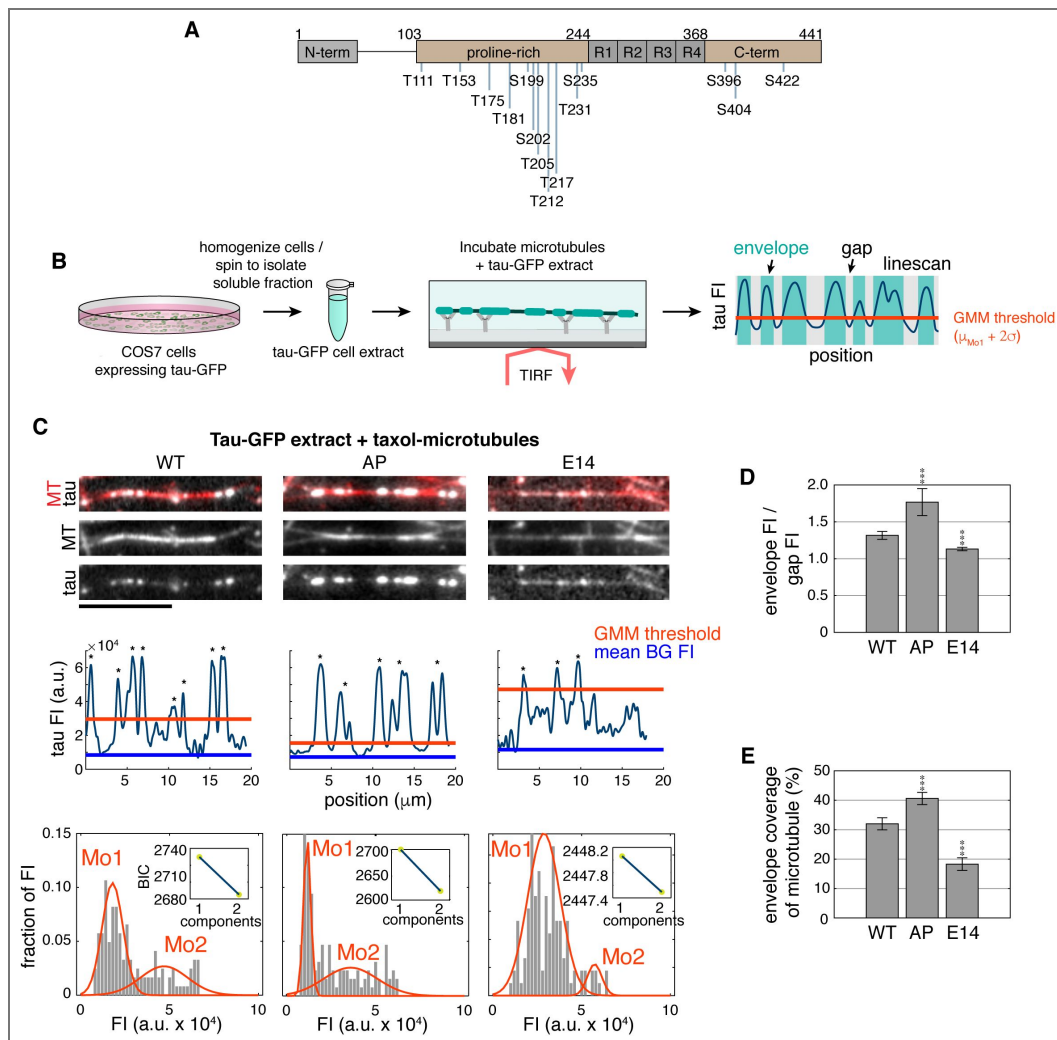


Figure 1. Tau hyperphosphorylation results in more diffuse microtubule interactions and reduces tau envelope formation in vitro.

A) Structure of 4R0N tau highlighting 14 disease-associated S/T residues mutated to E in pseudo-hyperphosphorylated (E14) tau or to A in phospho-resistant (AP) tau (Hoover et al., 2010). **B**) Schematic illustrating the main steps involved in preparing tau-GFP containing cell extracts and performing reconstitution assays with 500 nM tau (see Table 1). Linescans of tau intensity along a microtubule were analyzed using a Gaussian Mixture Model (GMM) to define intensity thresholds distinguishing envelopes from gaps. **C**) Representative images of WT, AP, and E14 tau-GFP on taxol-stabilized microtubules (WT: $n=183$, AP: $n=170$, E14: $n=170$). The indicated n values represent total number of samples over 3–4 replicates. Below the images, corresponding tau intensity plots along microtubules are shown, with horizontal lines indicating the GMM threshold (orange) and mean background fluorescence intensity (blue). Asterisks mark peaks identified as envelopes. Histograms show the distributions of tau fluorescence intensity fitted with a GMM. Insets show Bayesian Information Criterion (BIC) analyses used to determine whether a unimodal or multimodal distribution best describes the data. **D and E**) Plots quantifying the effects of hyperphosphorylation on D) tau envelope intensity and E) percentage of microtubule length covered by envelopes. Error bars indicate 95% CI. Statistical significance was assessed using Student's t-test (** $p < 0.0001$). Scale bars are 10 μm .

microtubules are irreversibly expanded, preventing tau from inducing lattice compaction and thus cooperative binding (Tan et al., 2019 [DOI](#); Siahaan et al., 2022 [DOI](#)). As expected, tau did not cooperatively bind to irreversibly expanded GMPCPP-microtubules and form significant envelopes on them, and overall tau intensity was lower than on taxol-stabilized microtubules (Fig S1A–C). E14 tau displayed significantly reduced intensities compared to WT and AP tau, which remained diffusely distributed but at higher intensities (Fig S1C). Together, these results indicate that phosphorylation influences not only cooperative binding, but also directly modulates tau's association with microtubules, independent of cooperativity. Phosphorylation likely also regulates tau-induced lattice compaction, in which misregulation of this process could have downstream effects on microtubule stability, as well as interactions with other MAPs and motor proteins.

Tau hyperphosphorylation enhances diffusive microtubule interactions and reduces the formation of envelopes in live neurons

Despite growing evidence for mechanisms that drive tau envelope formation *in vitro*, it remains unclear whether these structures form in native neuronal environments, what roles they play in regulating the microtubule cytoskeleton and intracellular transport, or how their formation is influenced by tau misregulation. To address these questions, we first sought to determine whether tau forms envelopes in neurons. Given the challenges associated with the expression of multiple tau isoforms in neurons and the variable phosphorylation of endogenous tau—both of which have a significant impact on tau-microtubule interactions—we employed a rescue approach by overexpressing WT, AP, or E14 tau-GFP in CRISPR-edited tau knockout (MAPT-KO) iPSC-derived neurons at DIV 7–8, when tau is normally expressed within the axon. This approach enabled controlled expression of a single tau isoform with specific phospho-mutations, which allowed us to isolate the effects of hyperphosphorylation without interference from the variable phosphorylation levels and heterogeneous mixture of tau isoforms normally expressed in neurons. We observed the formation of well-defined envelopes in axons, which were often enriched in discrete segments several microns in length (Fig. 2A [DOI](#)). To quantify the impact of hyperphosphorylation on envelope formation in neurons, we applied the same GMM-based thresholding method used in our *in vitro* studies (Fig. 2A [DOI](#)). Quantification showed that WT and AP tau formed envelopes of comparable intensity and width, whereas E14 tau envelopes were less intense and narrower (Fig. 2B, C [DOI](#)). Envelope distribution varied across conditions. AP tau exhibited a higher number of envelopes within discrete regions and a greater number of envelope-positive regions along axons compared to WT tau (Fig. 2D, E [DOI](#)). In contrast, E14 tau had the lowest occurrence of envelopes within discrete regions, and envelope-positive regions were less frequently observed throughout axons (Fig. 2D, E [DOI](#)). Together, these results suggest that phosphorylation modulates tau's cooperative binding to axonal microtubules, potentially influencing its role as a gatekeeper that controls access for other MAPs and motor proteins. Hyperphosphorylation may weaken this regulatory function, thereby altering motor engagement and contributing to axonal transport defects.

To further characterize the effects of tau hyperphosphorylation in cells, we performed fluorescence recovery after photobleaching (FRAP) assays to measure tau kinetics on microtubules. MAPT-KO neurons expressing WT, AP, or E14 tau-GFP were photobleached in a diffraction-limited region, and fluorescence recovery was monitored (Fig 2F [DOI](#)). AP tau recovered more slowly than WT, while E14 tau recovered the fastest, indicating that hyperphosphorylated tau is less stably bound to microtubules. The 1D geometry of axons constrains diffusion, so we repeated FRAP in COS-7 cells, which have a flatter, more 2D-like geometry. Recovery was ~2× faster in COS-7 cells (Fig S2). Furthermore, in COS-7 cells, WT and AP tau showed clear microtubule enrichment as the fluorescence signal along microtubules recovered above cytosolic levels, while E14 tau showed no microtubule enrichment, consistent with our *in vitro* data showing a lower ratio of E14 tau bound to microtubules than in solution (Fig S1C). These findings suggest that hyperphosphorylation disrupts tau's cooperative microtubule binding, increasing its exchange

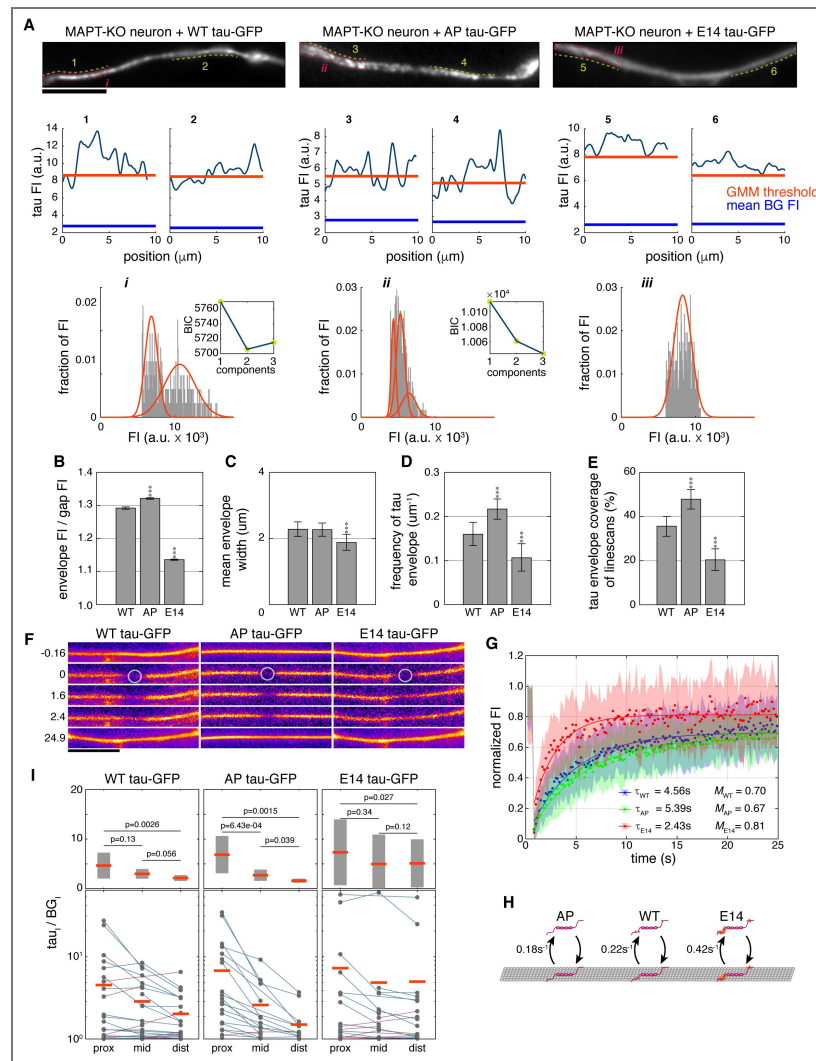


Figure 2. Tau hyperphosphorylation reduces the formation of envelopes in live neurons.

A) Representative images of axons with tau envelopes present in iPSC-derived MAPT-KO neurons expressing WT ($n=39$), AP ($n=39$), or E14 ($n=28$) tau-GFP. Each image is a max projection of 10 frames from live timelapse imaging. Images are all oriented so that the soma is towards the left and the distal axon is towards the right. Below, corresponding tau intensity plots from discrete locations (labeled 1–6) are shown. Horizontal lines indicating the GMM threshold (orange) and mean background fluorescence intensity (blue). Histograms of tau fluorescence intensity, fitted with a GMM to define the envelope threshold, are displayed for selected areas (labeled i–iii) marked by magenta ROIs. Insets show BIC analyses for multimodal distributions. **B–C**) Bar plots comparing B) tau envelope intensity and C) the mean envelope width of WT, AP, and E14 tau. **D–E**) Bar plots quantifying tau envelope frequency within D) discrete locations and E) the percentage of total length of axons imaged containing tau envelopes. Error bars indicate 95% CI. **F**) Timelapse images of fluorescence recovery after photobleaching assays of MAPT-KO neurons expressing WT ($n=20$), AP ($n=21$), or E14 ($n=20$) tau-GFP. Circles indicate the bleached regions. Note that tau envelopes are not clearly evident due to shorter exposure times needed for FRAP experiments and due to selection of higher tau-GFP expressing cells to ensure measurable recovery dynamics. **G**) Fluorescence recovery curves of WT (blue), AP (green), and E14 tau (red). Shaded regions indicate SD. The characteristic recovery (1) and mobile fraction (M) are indicated on the plot for each tau construct. **H**) Schematic illustrates how tau phosphorylation influences its dissociation from microtubules. Hyperphosphorylated E14 tau dissociates more readily than WT tau or the phospho-resistant AP tau, which remain more stably bound (Fig S2). **I**) Plots show how the ratio of tau signal in axons of neurons expressing WT ($n=37$), AP ($n=32$), and E14 ($n=36$) tau over background intensity varies for each tau construct across the proximal, mid, and distal axonal regions. In the bottom plot, blue lines indicate a decrease in the signal from the proximal towards the distal axon, and red lines indicate an increase in tau signal towards the distal axon. The top plot shows the means and grey bars represent SD. Orange bars show means. Wilcoxon signed rank test was used to determine the pairwise comparison of tau intensity between each axonal region as shown and the p-values are indicated in the above inset. The scale bars are 10 μm (A) and 5 μm (F). (* $p < 0.05$, ** $p < 0.001$, *** $p < 0.0001$).

between microtubules and the cytosol (Fig 2H [↗](#)). Thus, phospho-resistant AP tau forms stable envelopes with lower turnover than WT tau, whereas hyperphosphorylated tau binds microtubules diffusively and fails to form envelopes.

Tau's affinity for microtubules might also affect its distribution along the axon. We did a pair-wise comparison of the mean ratio of tau intensity over background to test how phosphorylation impacts the distribution of tau across the proximal, mid, and distal axonal regions. We found that WT and AP tau are more enriched proximally but E14 tau is more uniformly distributed along the axon (Fig 2I [↗](#)). These results suggest that hyperphosphorylation causes tau to mislocalize, which could dysregulate microtubules and transport in distinct axonal regions.

Tau phosphorylation differentially regulates kinesin-1 and kinesin-3 motility along microtubules

Kinesin-1 and kinesin-3 drive the anterograde transport of endosomes, lysosomes, mRNA, and other cargoes (Beaudet et al., 2024 [↗](#); Nagpal et al., 2024 [↗](#)). To determine how tau phosphorylation influences motor behavior, we analyzed its effect on individual kinesin motility using single-molecule TIRF assays. We analyzed the motility of Halo-tagged KIF5C (kinesin-1) and KIF1A (kinesin-3), expressed in COS-7 cells (Budaitis et al., 2022 [↗](#)), on immobilized taxol-stabilized microtubules in the presence of extracts from mock-transfected cells or from cells expressing WT, AP, or E14 tau. The use of mammalian-expressed motors and MAPs provided better protein quality and physiologically relevant PTMs. Results show that KIF5C exhibited increased pausing in the presence of WT, AP, and E14 tau compared to mock extracts without tau (Fig 3A, B [↗](#)). Run lengths and velocities were all reduced compared to mock, with AP tau having a slightly stronger reduction on run lengths compared to E14 tau (Fig 3E [↗](#), S3A and C). In contrast to KIF5C, tau significantly reduced KIF1A run frequency and processive motility (Fig. 3C, D [↗](#)). Analysis of KIF1A run lengths showed differences among constructs, with E14 tau being more inhibitory than WT or AP, while mean velocities remained unchanged (Fig. 3F [↗](#), S3D and F).

We next examined how kinesins behave inside versus outside of tau envelopes by measuring attachment and detachment frequencies. Tau envelopes were identified using the GMM method described above (Fig. 1C [↗](#)), and each kinesin trajectory was classified based on whether it started or ended inside or outside of an envelope (Fig. 3G [↗](#)). The attachment frequency of KIF5C did not significantly differ in the presence of WT, AP, or E14 tau compared to mock conditions, and did not differ significantly inside versus outside tau envelopes for any construct (Fig. 3H [↗](#)). However, KIF5C detached more frequently within envelopes containing AP tau compared to WT or E14 tau (Fig. 3H [↗](#)). In contrast, KIF1A showed a ~10-fold reduction in the attachment frequency with all tau constructs relative to mock lysate. Attachment was slightly elevated outside of AP tau envelopes compared to E14, while detachment occurred more frequently within E14 tau envelopes than outside (Fig. 3H [↗](#)). Together, these observations suggest that KIF1A is more inhibited by diffusely bound, non-cooperative tau on microtubules. Dissociation rate constants estimated from KIF5C and KIF1A dwell times also revealed significant differences in the presence of WT, AP, or E14 tau (Fig. S3B and E). KIF5C dissociation rate was slower in the presence of E14 tau compared to WT or AP tau, resembling its behavior on tau-free microtubules. Whereas KIF1A dissociation rate increased when exposed to E14 tau compared to WT or AP tau (Fig. 3I [↗](#)). These findings support the model that tau acts as a selective barrier to motor proteins, influencing cargo transport based on motor sensitivity. The differential inhibition of kinesin-1 and kinesin-3 suggests tau may gate transport by selectively modulating specific motors. Importantly, tau hyperphosphorylation appears to weaken its inhibitory effect on kinesin-1 while maintaining or enhancing its suppression of kinesin-3, potentially contributing to cargo transport deficits in neurodegenerative disease.

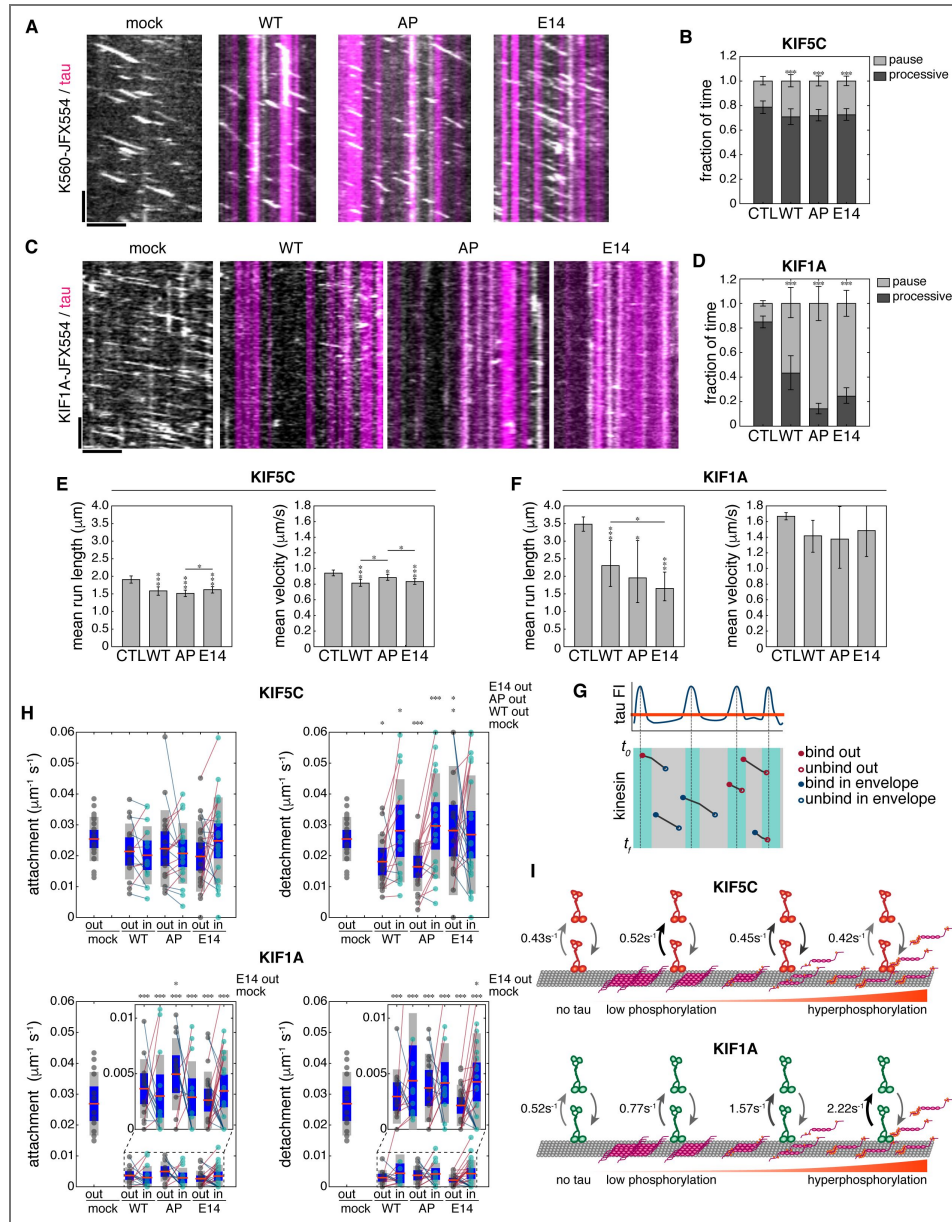


Figure 3. Tau phosphorylation differentially regulates kinesin-1 and kinesin-3 motility.

A and C) Kymographs of constitutively active Janelia Fluor 554 (JFX554)-labelled A) KIF5C(1–560)-JFX554 and C) KIF1A(1–393)-JFX554 motors moving along taxol-stabilized microtubules incubated with mock cell lysate or lysates containing 500 nM WT, AP, or E14 tau-GFP (magenta). KIF5C (mock: $n=1016$, WT: $n=792$, AP: $n=1078$, E14: $n=1203$) KIF1A (mock: $n=864$, WT: $n=173$, AP: $n=218$, E14: $n=305$) over 3–4 replicates. **B and D**) Bar graphs show the fraction of time that kinesin motors are paused or exhibit processive motility. Error bars indicate 95% CI. **E–F**) Bar graph shows the bootstrapped mean run length and mean velocity of E) KIF5C and F) KIF1A $\pm$ WT, AP, E14 tau-GFP. Error bars indicate SEM. **G**) Schematic illustrating the approach used to assess the impact of tau on kinesin-microtubule attachment and detachment. Motors were observed attaching to or detaching from microtubules either within tau envelopes or outside of them. **H**) Paired sample plots show the attachment and detachment frequencies of KIF5C (top) and KIF1A (bottom). Red lines indicate increased frequency inside envelopes compared to outside and blue lines indicate decreased frequencies inside of envelopes. Insets for KIF1A show a zoomed-in view of the attachment and detachment frequencies. Blue bars indicate 95% CI, orange lines mark mean values, and grey bars denote SD. **I**) Schematic illustrating the impact of tau on the detachment kinetics of kinesin motors. KIF5C is weakly inhibited by WT and phospho-resistant tau compared to hyperphosphorylated tau that has a similar dissociation rate compared to KIF5C in mock conditions. Conversely, KIF1A is more strongly inhibited by tau hyperphosphorylation and dissociates at a faster rate compared to WT and AP tau. (* $p < 0.05$, ** $p < 0.001$, *** $p < 0.0001$). Horizontal scale bars are 5 μm , vertical scale bars are 5 sec.

Hyperphosphorylation relieves the inhibition of lysosome motility by tau in neurons

We next examined the effects of tau hyperphosphorylation on lysosome transport in neurons. We labelled endogenous lysosomes using LysoTracker and monitored transport across proximal (~50 μm from the soma, avoiding the axon initial segment), mid (~midpoint), and distal (~50 μm from the axon terminal, avoiding the growth cone) axonal regions in control, MAPT-KO, and MAPT-KO neurons expressing phospho-mutant tau-GFP constructs at DIV 7–8 (Fig. 4A). Kymographs were generated from timelapse images, and lysosome trajectories were analyzed using KymoButler followed by custom tracking analysis scripts. Lysosomal transport was dominated by short-range, diffusive motility, with long-range transport events accounting for only a small fraction of total lysosome movement. In control neurons, ~5% of lysosomes travelled longer than 10 μm with fewer trajectories exceeding 20 μm (Fig. 4B, C). MAPT-KO neurons exhibited an increased fraction of retrograde biased long-distance travel. Whereas the fraction of lysosomes travelling long distances in WT tau-expressing MAPT-KO neurons resembled controls. However, expression of AP tau reduced anterograde and retrograde long-range travel, while E14 tau increased long-range travel in both directions (Fig. 4C). Although we observed changes in the frequency of long-range lysosome transport, the mean displacement across all trajectories was less affected. E14 tau had the strongest effect in the proximal and distal axon, increasing the mean displacements and cell-to-cell variability relative to WT and AP tau (Fig. 4D).

To assess how the effects on motility varied as a function of tau expression, we classified neurons by expression level using the ratio of the mean tau-GFP pixel intensity in the axon relative to mean background pixel intensity. The effects on lysosome travel distances were dependent on the level of tau expression and scaled differently with each construct. In WT tau expressing MAPT-KO neurons, lysosomes less frequently traveled long distances at high expression levels ($> 5\times$ background signal), compared to those in neurons with lower (1.1–2.0 $\times$) or mid-range (2.0–5.0 $\times$) tau expression (Fig. S4). In MAPT-KO neurons expressing AP tau, the reduction in lysosome transport distance scaled more steeply with tau intensities, with long travel distances strongly reduced even at lower expression levels (Fig. S4). Conversely, in E14 tau-expressing MAPT-KO neurons, long-range lysosome transport was enhanced at low intensities but reduced at high intensities, although to a lesser extent than in neurons expressing WT or AP tau (Fig. S4). We also observed that MAPT-KO neurons with or without expression of tau-GFP had altered lysosome distributions along the axon. In control neurons, lysosomes were most abundant in the proximal axon relative to mid and distal regions (Fig. 4E). In MAPT-KO neurons, lysosomes were as abundant in the mid axon and displayed increased cell-to-cell variance compared to controls (Fig. 4E). WT tau expression in MAPT-KO neurons partially restored lysosome distribution, with lysosome abundance comparable in the mid and distal regions, but slightly reduced in the proximal axon relative to controls. AP tau-expressing MAPT-KO neurons, however, had significantly fewer lysosomes in the proximal axon and showed reduced variability compared to WT. In contrast, E14 tau-expressing MAPT-KO neurons displayed similar lysosome abundance across all axonal regions but increased cell-to-cell variance in the proximal axon (Fig. 4E). These data suggest that lysosomes are normally enriched in the proximal axon and that tau promotes lysosome flux toward this region. It is important to note that iPSC-derived neurons at DIV 7–8 have been shown to predominantly express 3R tau isoforms (Iovino et al., 2010; Iovino et al., 2015; Sposito et al., 2015). In our rescue experiments, however, we introduce 4R0N tau. Although we observed minimal differences in displacements and lysosome abundance between MAPT-KO neurons expressing WT (4R) tau and control neurons (3R), there is still the possibility that isoform differences influence the severity of transport phenotypes, particularly under conditions of increased phosphorylation.

Lysosomes alternate between periods of stationary, diffusive, and processive movement. Trajectories were segmented by identifying directional reversal events and the intervening periods were characterized as stationary (less than 0.16 μm , within the tracking uncertainty), diffusive (between 0.16 μm and 1.2 μm), and processive (greater than 1.2 μm) (Fig. 5A). In control

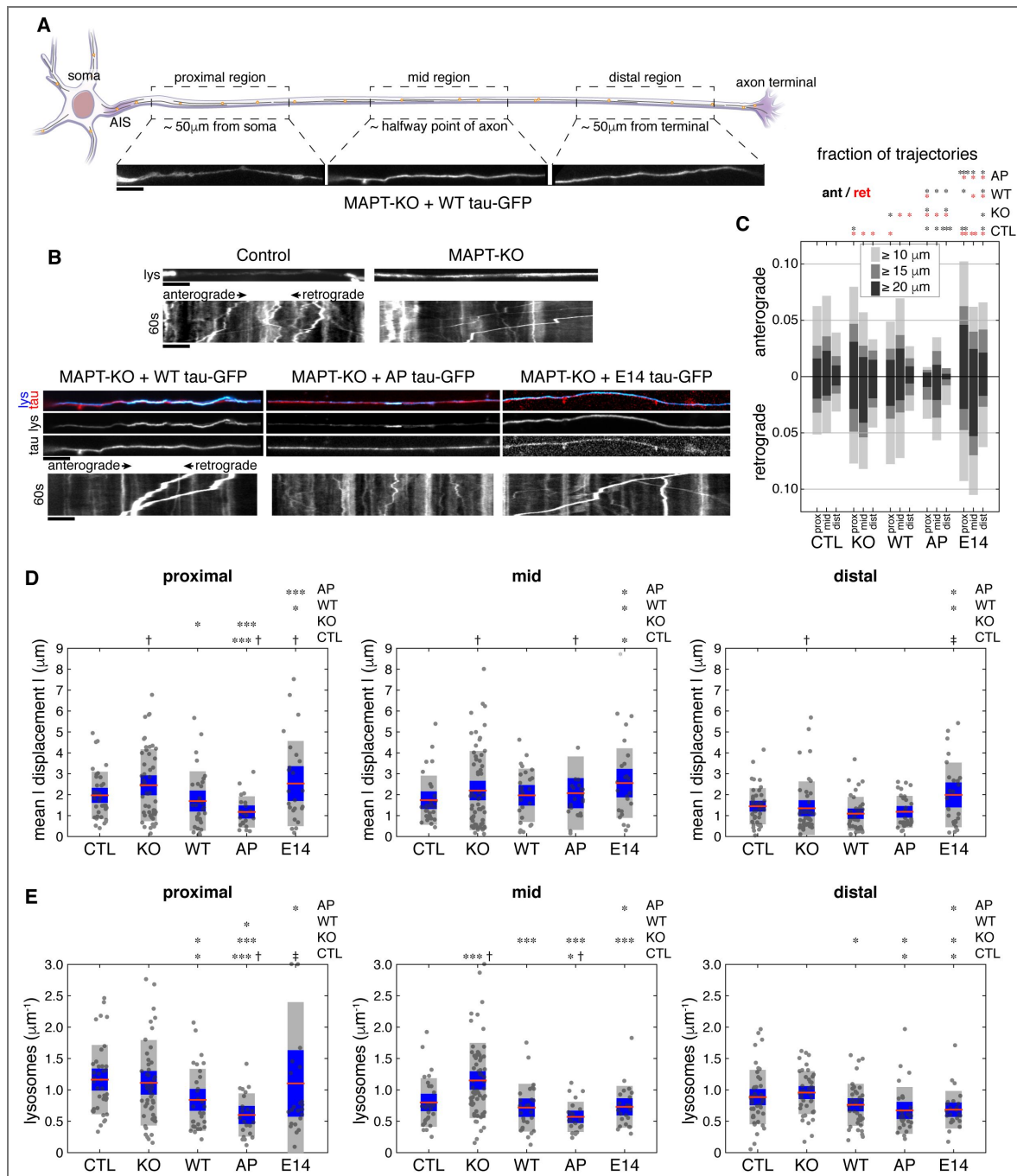


Figure 4. Tau perturbations impact the trafficking of lysosomes in neurons.

A) Schematic of a neuron indicating the regions where tau's effects on lysosome transport were assessed. Lysosome transport was recorded in proximal, mid, and distal axonal regions. Example images of WT tau-GFP expression are shown from each region. Proximal regions were defined as ~50 μm from the soma, mid-axonal regions as ~halfway along the axon, and distal regions as ~50 μm from the axon terminal. Images are all oriented so that the soma is towards the left and the distal axon is towards the right. The number of cells analyzed under each condition are indicated in Table 2. **B)** Images show mid-axonal tau-GFP signal and max projections of lysosomes (lys) in control, MAPT-KO, and MAPT-KO neurons expressing WT, AP, and E14 tau-GFP. Below kymographs show anterograde and retrograde lysosome transport for each condition. **C)** Bar plot shows the fraction of anterograde and retrograde long-distance trajectories for control (CTL), MAPT-KO (KO), and MAPT-KO neurons expressing WT, AP, or E14 tau-GFP. Above the plot, asterisks represent statistical significance tested for anterograde (black) and retrograde (red) trajectories (* $p < 0.05$, ** $p < 0.001$, *** $p < 0.0001$). **D and E)** Plots show the D) mean absolute displacements and the E) frequency of lysosomes in the proximal, mid, and distal axon for each condition. Blue bars indicate 95% CI, orange lines mark mean values, and grey bars denote SD. Statistical significance is indicated by asterisks above the at the top of plots (* $p < 0.05$, ** $p < 0.001$, *** $p < 0.0001$). Sample variance significance was determined using a two-sample F -test for equal variances ($^\dagger p < 0.05$, $^\ddagger p < 0.001$). Scale bars are 10 μm .

neurons, lysosomes were primarily in diffusive or stationary states, and exhibited processive movement only a small fraction of the total time— a trend consistent across all regions (Fig 5A, B). MAPT-KO neurons exhibited an increase in stationary lysosome behavior and a corresponding decrease in processive movement approaching the distal axon. In MAPT-KO neurons expressing AP tau, lysosomes exhibited increased diffusive behavior and in certain cases were less processive (Fig 5B). Although these data did not reach statistical significance, when averaged across trajectories and cells, the data suggest a trend of stronger effects in the proximal axon, with E14 tau increasing processive time, AP tau reducing it, and WT tau exhibiting an intermediate effect. Lysosomes are bidirectional cargoes that frequently undergo directional reversals. MAPT-KO neurons exhibited an increased frequency of directional reversals in the proximal and mid axon compared to control neurons (Fig 5C). AP tau significantly reduced the reversal frequency, while E14 tau enhanced reversals, with the most significant effects observed in the proximal and mid axon (Fig 5C). These data suggest that non-phosphorylated AP tau promotes more diffusive or stationary transport modes, whereas E14 tau may restore aspects of processivity but impairs directional persistence. An important consideration is that while these measurements capture a snapshot of lysosome dynamics, the cumulative effects of altered transport and distribution over longer timescales are likely to be more pronounced.

The effects of tau on individual kinesin and dynein motors have been well characterized in single-molecule motility assays, but it is less clear how tau effects endogenous cargoes such as lysosomes or other endocytic vesicles that are transported by multiple motor types and variable motor copy numbers (Jordens et al., 2001; Brown et al., 2005; Loubéry et al., 2008; Rosa-Ferreira and Munro, 2011; Bentley et al., 2015; Pu et al., 2016; Beaudet et al., 2024). To better understand how tau hyperphosphorylation affects the coordinated activity of motor teams driving bidirectional transport, we analyzed the processive segments of lysosome motility. In most neurons analyzed, lysosomes showed a slight retrograde bias (Fig 5D). Although lysosomes in MAPT-KO neurons spent less time in the processive state overall, processive runs exhibited comparable anterograde run lengths but significantly longer retrograde run lengths compared to control neurons (Fig 5E). Additionally, mean velocities in both anterograde and retrograde directions were increased compared to control neurons (Fig 5F). In MAPT-KO neurons expressing WT tau, processive lysosome motility closely resembled that of control neurons, with comparable run lengths and velocities in both directions across most axonal regions (Fig. 5E, F). AP and E14 tau expression in MAPT-KO neurons had distinct effects on processive motility. AP tau generally suppressed anterograde processive transport while having weak or no effect on retrograde transport (Fig. 5E, F). Conversely, E14 tau either rescued to WT-like transport characteristics or in most regions enhanced lysosome run lengths and velocities.

These findings support a model in which non-phosphorylated tau suppresses motor-driven transport, whereas hyperphosphorylated tau alleviates this inhibition and functionally mimics tau deletion (Fig. 5G). Due to its proximal enrichment and tighter microtubule binding along the axon (Fig. 2H, I), AP tau strongly inhibits both anterograde and retrograde transport in the proximal region, with reduced inhibition of retrograde movement in the mid and distal axon. In contrast, E14 tau, which binds microtubules more weakly and distributes more uniformly along the axon (Fig. 2H, I), relieves inhibition and, in many cases, enhances both anterograde and retrograde motility across axonal compartments. Collectively, these data support a model in which tau phosphorylation alters microtubule affinity, leading to changes in the axonal accumulation and regional concentrations of tau. These alterations modulate tau's capacity to act as a roadblock and influence its effect on specific motor proteins driving axonal transport.

Discussion

Tau hyperphosphorylation is a hallmark of Alzheimer's Disease and associated with the aggregation of tau into neurofibrillary tangles. Here, we focused on the functional consequences of tau hyperphosphorylation on microtubules—either as a cytoskeletal organizer or as a regulator of intracellular transport— prior to aggregation. We find that hyperphosphorylation disrupts tau's ability to form envelopes on microtubules and alters microtubule-based transport. Tau binds

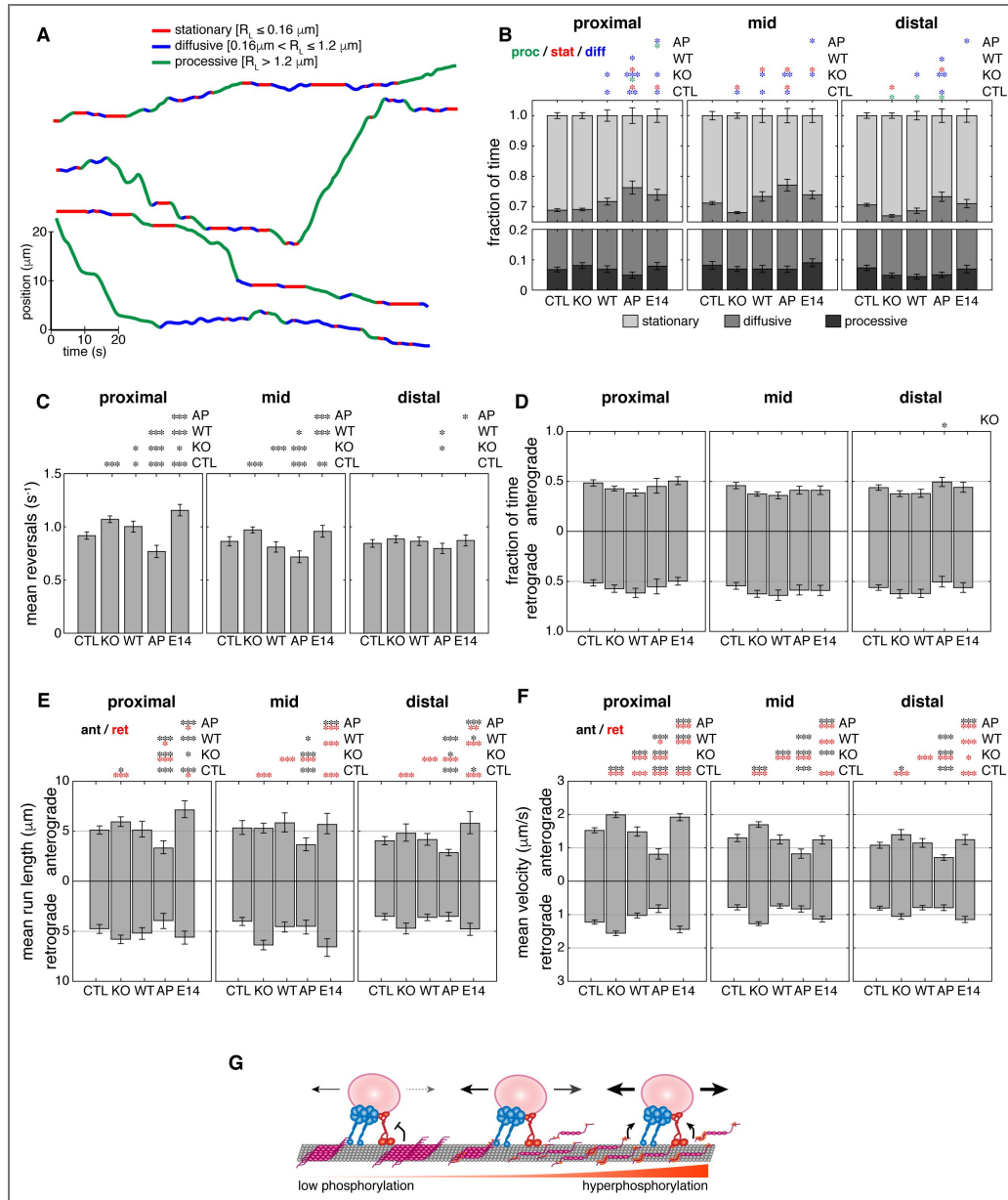


Figure 5. Tau hyperphosphorylation relieves inhibition of processive lysosome motility in neurons.

A) Shown are examples of lysosome trajectories segmented into stationary (red), diffusive (blue), and processive (green) motility based on run length. Stationary segments were defined as periods between two reversal events with a run length (R_L) $< 0.16 \mu\text{m}$; diffusive segments as those with run lengths between $0.16 \mu\text{m}$ and $1.2 \mu\text{m}$; and processive segments as those $> 1.2 \mu\text{m}$. **B**) Plots show the fraction of time spent in each transport mode for control (CTL), MAPT-KO (KO), and MAPT-KO neurons expressing WT, AP, and E14 tau-GFP in the proximal, mid, and distal axon. Error bars represent 95% CI. Statistical significance is indicated above the plots for comparison of the fraction of processive (proc; green) diffusive (diff; blue), and stationary (stat; red) time for each condition. **C–D**) Bar plots show **C**) the mean reversal frequency of lysosomes, **D**) and the fraction of time of anterograde or retrograde directed processive motility. **E–F**) Bar plots show **E**) the mean run lengths and **F**) mean velocities of processive runs of lysosomes for each condition in proximal, mid, and distal axonal regions. Error bars in **C**), **E**), and **F**) indicate SEM and in **D**) indicate 95% CI. The number of trajectories and cells analyzed under each condition are indicated in Table 2. **G**) Schematic summarizing the impact of tau hyperphosphorylation on bidirectional lysosome transport in neurons. Low levels of tau phosphorylation reduce processive anterograde transport throughout the axon but has a weaker impact on retrograde transport, whereas hyperphosphorylated tau enhances lysosome motility in both directions across the axon, similar to the effects observed in tau knockout conditions. Statistical significance is shown above each plot, where red asterisks indicate comparisons of retrograde transport and black asterisks indicate comparisons of anterograde transport. (* $p < 0.05$, ** $p < 0.001$, *** $p < 0.0001$).

cooperatively in live neurons, consistent with its behaviour on stabilized microtubules *in vitro* (Figs. 1 [↗](#) and 2 [↗](#)). Disease-related hyperphosphorylation ablates cooperative binding and enhances tau's dissociation from microtubules (Figs. 1 [↗](#) and 2 [↗](#)). Tau hyperphosphorylation also has different effects on different kinesin motors. KIF5C motility is mildly inhibited by mammalian expressed tau compared to KIF1A that is strongly inhibited (Fig. 3 [↗](#)). Hyperphosphorylation reduces KIF5C dissociation from microtubules, while it increases that of KIF1A (Fig. 3 [↗](#)), suggesting that misregulation of tau could alter axonal trafficking by disrupting kinesin activity. In agreement, we found that hyperphosphorylation weakens tau-mediated inhibition of lysosome transport in neurons. While unphosphorylated tau acts as a strong barrier to anterograde lysosome movement, hyperphosphorylated tau alleviates this inhibition, mimicking tau knockout conditions in which lysosomes moved more processively (Figs. 4 [↗](#) and 5 [↗](#)). These disruptions in transport likely lead to impairments of degradative pathways and protein homeostasis, which contribute to neuronal dysfunction and degeneration in tauopathies.

Isolating the effects of tau phosphorylation presents significant challenges. Tau is phosphorylated at numerous residues and at different developmental stages, and individual tau molecules often exhibit heterogeneous phosphorylation patterns (Lindwall and Cole, 1984 [↗](#); Mandelkow et al., 1995 [↗](#); Trinczek et al., 1995 [↗](#); Hasegawa et al., 1998 [↗](#); Siahaan et al., 2024 [↗](#); Fan et al., 2025 [↗](#)). There are six tau isoforms expressed in the mature human brain (Weingarten et al., 1975 [↗](#); Goedert et al., 1988 [↗](#); Goedert et al., 1989a [↗](#); Goedert et al., 1989b [↗](#); Andreadis et al., 1992 [↗](#)), further complicating efforts to dissect specific aspects of its function and regulation. To isolate effects of tau phosphorylation, we employed phospho-mimetic and phospho-resistive mutants of tau (Hoover et al., 2010 [↗](#)). To avoid effects from various endogenous tau isoforms, we used either COS-7 cells to express tau-GFP for *in vitro* studies, since tau is not endogenously expressed in these cells, or expressed the phosphomutant tau proteins in MAPT-KO neurons to assess transport phenotypes. These systems enabled us to examine tau function in a near-physiological environment—preserving relevant cytosolic factors and PTMs—while isolating the effects of hyperphosphorylation of a single tau isoform from the multiple isoforms of endogenous tau. While this study focused on one isoform, other tau isoforms differ in microtubule affinity and likely undergo distinct phosphorylation patterns. Typically, iPSC-derived neurons at DIV 7–8 predominantly express 3R tau, with 4R tau appearing later (Iovino et al., 2010 [↗](#); Iovino et al., 2015 [↗](#); Sposito et al., 2015 [↗](#)). Since 3R tau has a lower affinity for microtubules but more strongly inhibits kinesin and dynein motors compared to 4R tau (Vershinin et al., 2008 [↗](#); McVicker et al., 2011 [↗](#); McVicker et al., 2014 [↗](#)), hyperphosphorylation may have more severe effects in mature neurons where 3R and 4R tau are equally expressed. Future work is needed to explore how these isoforms interact, influence each other's microtubule binding and cooperative assembly, and how phosphorylation heterogeneity among tau molecules (Siahaan et al., 2024 [↗](#); Fan et al., 2025 [↗](#)) contributes to disease progression in tauopathies.

Our findings contribute to the growing body of evidence that the N-terminal projection domain of tau modulates cooperative binding to microtubules (Tan et al., 2019 [↗](#); Siahaan et al., 2019 [↗](#); Siahaan et al., 2022 [↗](#)), and that mutations or modifications in this region impair tau's ability to form envelopes (Cario et al., 2022 [↗](#); Siahaan et al., 2024 [↗](#)) and function as an effective regulator of intracellular transport (Kanaan et al., 2012 [↗](#); Stern et al., 2017 [↗](#); Balabanian et al., 2022 [↗](#)). Previous studies show that tau binds to compacted GDP-lattices or cooperatively binds and compacts taxol-stabilized microtubule lattices, but cannot bind cooperatively to GMPCPP-microtubules (Tan et al., 2019 [↗](#)). Consistent with this, we observed robust envelope formation on taxol-microtubules but not on GMPCPP-microtubules (Figs. 1 [↗](#) and S1), indicating that lattice compaction is required for cooperative binding. The projection domain likely facilitates this process either by modulating the microtubule-binding repeats or by promoting tau–tau interactions that drive cooperative binding. Supporting the latter model, the pathogenic R5L mutation was shown to disrupt cooperative binding by altering the structure of the projection domain without affecting the microtubule-binding repeats (Cario et al., 2022 [↗](#)), implicating the projection domain as a key mediator of tau–tau interactions. Consistent with this, our data show that cooperative binding via the projection domain is essential for maximizing tau association with microtubules (Fig. 1 [↗](#)). Similarly, we found that phosphorylation at sites primarily within

the projection domain significantly reduced cooperative binding, diminished envelope formation, and increased tau dissociation from microtubules, indicating weakened binding affinity (Fig. 2F–G, Fig. S2).

If phosphorylation within this region strictly regulates cooperative binding, we would expect no effect of hyperphosphorylation on tau binding to GMPCPP-microtubules, where cooperative binding is absent. However, this is not the case, since we found that hyperphosphorylated tau exhibited significantly reduced binding to GMPCPP-microtubules compared to WT and phospho-resistant tau (Fig. 1G–I). This suggests several possibilities: (1) the projection domain may allosterically regulate the microtubule-binding repeats to control direct microtubule interactions, (2) phosphorylation within the projection domain and C-terminus may reduce the intrinsic affinity of tau for the microtubule lattice, or (3) phosphorylation may impair tau's ability to compact the underlying lattice, thereby preventing envelope formation. Taken together, our findings reconcile multiple lines of evidence to establish the projection domain as a critical regulator of both tau–tau and tau–microtubule interactions. Phosphorylation at sites within this region modulate these interactions and controls tau's binding dynamics. Disruption of this regulatory mechanism increases the volatility of tau–microtubule interactions, potentially contributing to microtubule destabilization and impaired intracellular transport.

Our data demonstrate that tau hyperphosphorylation is associated with impaired axonal transport. Previous single-molecule studies established that tau inhibits kinesin-1 and kinesin-3 motility, while having less of an effect on kinesin-2 and dynein. Based on these findings, conditions that reduce tau's microtubule association, such as hyperphosphorylation or tau knockout, would be expected to enhance anterograde transport of cargoes driven by kinesin-1 and kinesin-3 motors. Conversely, dephosphorylation, which strengthens tau's association with microtubules, would be expected to further inhibit anterograde transport. Consistent with this model, our results show that both tau knockout and tau hyperphosphorylation result in increased anterograde and retrograde motility. Whereas phospho-resistant tau reduced anterograde transport and had little-to-no impact on retrograde transport. These findings suggest that hyperphosphorylated tau fails to perform its normal function as a regulatory barrier on microtubules. The observed increase in retrograde transport may indicate that tau also modulates dynein activity, and that hyperphosphorylation either relieves tau-mediated inhibition of dynein or indirectly alters kinesin function in a way that favors retrograde transport. One explanation is that cargoes are transported by teams of multiple motors, including force-producing motors and those that act as tethers (Feng et al., 2018; Arpag et al., 2019). For instance, increased inhibition of kinesin-3 by hyperphosphorylated tau could reduce its tethering capacity, shifting the balance in favor of dynein-driven retrograde transport. This is consistent with the increased retrograde lysosome motility we observed in E14 tau-expressing MAPT-KO neurons. Alternatively, mislocalization of tau along the axon may permit increased binding of other MAPs, such as MAP7. MAP7 competes with tau for microtubule lattice binding and has been shown to enhance kinesin-1 motility while inhibiting kinesin-3 (Monroy et al., 2018). These opposing effects may contribute to enhanced anterograde transport by promoting kinesin-1 processivity or increased retrograde transport by limiting kinesin-3 activity, while having a comparatively minor impact on dynein-mediated transport. This suggests that cargo-specific motor compositions and their differential sensitivities to tau (Beaudet et al., 2024) may underlie the observed changes in transport dynamics when tau is hyperphosphorylated.

Taken together, our findings show that tauopathy-related hyperphosphorylation weakens tau's interaction with microtubules, reducing its ability to form cohesive envelopes and regulate axonal transport. Beyond impaired lysosome trafficking, this disruption likely affects signaling and other maintenance pathways essential for axonal health and function. Our results suggest a potential pathway, where prior to oligomerization and late-stage aggregation, hyperphosphorylated tau first impairs transport, which then contributes to axonal decline. As aggregation progresses, these effects are amplified, leading to further trafficking defects, axonal blockages, inflammation, and ultimately neurodegeneration. While our study highlights early phenotypes of tauopathy, how these changes interact with other disease mechanisms remains unclear. Future work should

define how tau misregulation reshapes the neuronal environment and cooperates with additional pathological processes to drive neurodegeneration. Overall, our results suggest that the disruption of axonal transport—caused by tau’s impaired association with microtubules—is one of the earliest drivers of neurodegeneration.

Methods

DNA constructs

The pRK5-EGFP-Tau (Addgene plasmid # 46904), pRK5-EGFP-Tau AP (Addgene plasmid # 46905), and pRK5-EGFP-Tau E14 (Addgene plasmid # 46904) plasmids were gifts from Karen Ashe. KIF1A(1-393)-LZ-HALO reported in (Budaitis et al., 2021) was generously provided by Kristen Verhey (U. Michigan). KIF5C(1-560)-HALO reported in (Twelvetrees et al., 2016) was generously provided by Erika Holzbaur (U. Pennsylvania).

Cell culture and transfection

COS-7 monkey kidney fibroblast cells (ATCC) were maintained in DMEM (Gibco), supplemented with 10% fetal bovine serum (Thermo Fisher Scientific) and 1% GlutaMAX (Gibco) and cultured in MatTek glass-bottom dishes (No. 1.5 Coverslip) (MatTek Corporation, Ashland, MA) at 37°C with 5% CO₂ for 24–48 h before transfection. Cells were then transfected with 500 ng of DNA plasmid prepared in OPTI-MEM Reduced Serum (Life Technologies), using Lipofectamine LTX with Plus-Reagent, according to the manufacturer’s instructions (Invitrogen, Thermo Fisher Scientific) for 24 h prior to imaging or homogenization.

Human neurons were induced from iPSC control and MAPT-KO AIW002-02 lines. The iPSCs were obtained from the C-BIG repository (Montreal Neurological Institute-Hospital, Montreal, QC, Canada) and were generated from the PBMC of a control 37-year-old male donor. MAPT knockout was achieved by CRISPR-Cas9 and verified by PCR and sequencing. Guide RNAs (gRNAs) were designed using “optimized CRISPR design” tool (www.crisp.mit.edu). The locations of gRNA1(GG ATAAGTTCTGAGGAGTGT) and gRNA2 (TGATCTGGGCCTGCTGTGCA) were around exon 2 with ATG start codon. Oligonucleotides with Bsb1 cleavage overhang were ordered from Life Technologies, annealed and cloned into the PX459 Cas9/puromycin vector (Addgene plasmid #48139). iPSCs were characterized to ensure correct genotyping, pluripotency, and screened for microbiology/virus. iPSCs were maintained in mTeSR Plus media (STEMCELL Technologies, Vancouver, BC) on Matrigel (Corning, Corning, NY) coated culture dishes using Gentle Cell Dissociation Reagent (STEMCELL Technologies) for passaging. The iPSCs were cultured for ~ 30 days in STEMdiff SMADi medium (STEMCELL Technologies) to induce the formation of neuronal progenitor cells (NPCs). NPCs were dissociated from culture dishes (Accutase Cell Dissociation Reagent) and seeded on poly-l-ornithine (PO) (10 µg/ml)/laminin (15 µg/ml)-coated MatTek glass-bottom dishes (No. 1.5 Coverslip) at 37°C with 5% CO₂ in BrainPhys differentiation media (STEMCELL Technologies) at about 2,000 cells per cm². Six days after seeding, the cells were transfected with lipofectamine LTX reagent at a 3:1 reagent:DNA according to the manufacturer’s instructions then imaged at 7–8 days following terminal differentiation induction.

Live cell imaging

To capture lysosome dynamics, neurons were treated with 70 nM LysoTracker (Thermo Fisher Scientific, Waltham, MA) for 30 minutes at 37°C with 5% CO₂ prior to replacing the media with BrainPhys containing 15 mM HEPES (H3375, Millipore Sigma). Live cell imaging was performed on an Eclipse Ti-E inverted microscope (Nikon) with custom optics for total internal reflection fluorescence (TIRF) and imaged using EMCCD camera (iXon U897; Andor Technology) maintained at 37°C. Time lapse recordings were acquired with 120 msec exposures using a 640 nm laser (100 mW) set at 1% power for 120 sec per cell using NIS-Elements acquisition software (Nikon). To image tau-GFP envelopes in neurons, imaging was performed as described but acquired with 500 msec exposures for 30 seconds using a 480nm laser set at 2% laser power. To measure tau-GFP expression in neurons a single frame was taken with 500nm exposure using a 480nm laser set at

2% laser power. Image files were exported as TIFFs, which were opened with ImageJ (NIH) and lysosomes were tracked by kymograph analysis. Briefly, linescans were traced over the axon to generate kymographs, which were then analyzed using KymoButler (Jakobs et al., 2019) to identify individual lysosome trajectories. The coordinates of each trajectory were then imported into MATLAB (The MathWorks) to analyze the mode of transport, directionality, displacement, and number of reversals for all trajectories within cells using custom scripts.

FRAP

COS-7 cells or neurons at 7–8 DIV expressing WT, AP, or E14 tau-GFP were used to assess tau kinetics in live cells. FRAP experiments were performed at 37 °C using a DeltaVision OMX system equipped with a 100×/1.40 NA objective lens and an EMCCD camera (Evolve, Photometrics). Prior to photobleaching, 5 pre-bleach images were acquired with 100 ms exposure. Photobleaching was performed within a diffraction limited spot using 50% 488 nm laser power for 50 ms. Fluorescence recovery was then recorded for 100 frames for COS-7 cells or 150 frames for neurons using 31% 488 nm laser power and 100 ms exposure per frame. Fluorescence recovery curves were generated using ImageJ, and background-corrected to account for acquisition-related photobleaching. Fluorescence recovery data were fit to a single-exponential function described by equation (1):

$$F(t) = M(1 - e^{-\frac{t}{\tau}}) \quad (1)$$

where $F(t)$ is the fluorescence intensity at time t , M is the mobile fraction, τ is the characteristic recovery time and the dissociation rate constant (k_{off}) is described by equation (1.1):

$$k_{off} = -\frac{1}{\tau} \quad (1.1)$$

COS-7 cell extract preparation

To prepare tau-GFP-containing cell extracts, COS-7 cells expressing WT, AP, or E14 tau-GFP were harvested 24 h post-transfection. For extracts containing kinesin motor proteins, COS-7 cells expressing Halo-tagged KIF5C or KIF1A were labeled with Janelia Fluor JFX554 ligand (Promega) using the manufacturer's rapid labeling protocol and collected 24 h post-transfection. Cells were incubated with 200 nM JFX554 for 30 min, rinsed 2× with pre-warmed DMEM, followed by 2× PBS to remove residual media. Cells were then collected using a cell scraper in ice-cold lysis buffer (10 mM PIPES, 50 mM potassium acetate, 4 mM MgCl₂, 1 mM EGTA, pH 7.0), supplemented with protease inhibitor cocktail (BioShop), 10 mM DTT, and 0.5% Triton X-100. For motor-containing extracts, 1 mM MgATP was also included. Cell suspensions were incubated on ice for 1 min, then centrifuged at 1,000 × g for 10 min at 4 °C. Supernatants were transferred to fresh ice-cold microcentrifuge tubes and spun at 16,000 × g for 10 min at 4 °C. The concentration of tau-GFP in the extracts was determined by measuring absorbance at 488 nm using a Nanodrop spectrophotometer, applying an extinction coefficient of 56,000 M⁻¹cm⁻¹ (Table 1). The resulting soluble fractions containing tau-GFP or fluorescently labeled kinesin motors were aliquoted, flash-frozen in liquid nitrogen, and stored at –80 °C for subsequent experiments.

Microtubule polymerization

To prepare taxol-stabilized microtubules, polymerization reaction mixtures containing 8% Alexa647-labeled tubulin and 92% unlabeled tubulin (final concentration of 5 mg/mL) in BRB80 (80 mM K-Pipes, 1 mM MgCl₂, 1 mM EGTA, pH 6.8) supplemented with 1 mM GTP were prepared on ice then incubated for 25 mins at 37°C. Microtubules were then stabilized with 20 μM Taxol (Cytoskeleton) and incubated for 25 mins at 37°C. For GMPCPP-microtubules, reaction mixtures were mixed with 1mM GMPCPP (Jena Bioscience, Jena, Germany) for 60 minutes. Microtubules were cleared 2X by pelleting them at 10,600g for 5 mins at RT then washed with T-BRB80 (BRB80 supplemented with 20 μM Taxol).

In vitro reconstitution assays

Flow chambers were first incubated with anti- β -tubulin (T4026 clone TUB2.1, Sigma) diluted 1:25 in BRB80 for 5 mins. Chambers were then treated with F-127 for 5 mins and washed 2X with T-BRB80. Fluorescently-labeled microtubules were added to the chambers and incubated for 5 mins at RT. Unbound microtubules were washed out with 2X T-BRB80. Tau-GFP containing cell extracts were diluted 2-fold in MAB supplemented with 0.2 mg/ml BSA, 10 mM DTT, 20 mM Taxol, 15 mg/ml glucose, ≥ 2000 units/g glucose oxidase, ≥ 6 units/g catalase, and 1 mg/ml casein and flown through the chamber containing microtubules and imaged in TIRF. To image tau-GFP envelopes in vitro, images were acquired with 500 msec exposures for 30 seconds using a 480nm laser set at 2% laser power. Envelope analysis was performed using images of sum projections of stacks consisting of 10 frames with custom MATLAB scripts.

To capture kinesin motility dynamics, motility reaction mixtures containing 3 μ l of KIF5C-JFX or KIF1A-JFX motor-containing cell extracts were mixed with tau-GFP cell extracts diluted 2-fold in MAB supplemented with 0.2 mg/ml BSA, 10 mM DTT, 20 mM Taxol, 15 mg/ml glucose, ≥ 2000 units/g glucose oxidase, ≥ 6 units/g catalase, 1 mg/ml casein, and 5mM MgATP and flown through the chamber containing microtubules and imaged in TIRF. To image kinesin motility, images were acquired for 2 minutes with 120 msec exposure using 640 nm laser set at 10–15% laser power. Image files were exported as TIFFs, linescans were traced over microtubule images to generate kymographs of kinesin motility, which were then analyzed using KymoButler and custom MATLAB scripts.

Envelope analysis

To analyze tau envelopes, linescans were traced along microtubules incubated with WT, AP, or E14 tau-GFP, and fluorescence intensity profiles were generated. Tau intensity distributions were then fit using a Gaussian mixture model (GMM) to determine the intensity threshold for defining envelopes. The optimal number of components in each distribution was selected based on the Bayesian Information Criterion (BIC). Microtubules with tau envelopes were identified by multi-component fits, whereas those lacking envelopes were best fit by a single-component model (Fig. 1C). The envelope detection threshold was defined as the intensity corresponding to the first GMM mode plus two standard deviations (σ). Intensities above this threshold were classified as envelopes, while those below were considered diffuse tau between envelopes. A similar analysis was performed to identify and quantify tau envelopes in neurons, with the exception that 10 μ m linescans were drawn in envelope-positive regions, which were manually selected based on the degree of fluorescence intensity variation along the axon.

Analysis of kinesin kinetics on tau-microtubules

To quantify the attachment and detachment frequencies of kinesins on tau-decorated microtubules, tau envelopes were first identified as described above. Kymograph analysis was then used to determine the start and stop positions of each kinesin trajectory, which were categorized as either inside or outside of tau envelopes (Fig. 3G). Pairwise comparisons were performed to assess the significance of differences in attachment and detachment frequencies inside versus outside of envelopes for each tau construct.

To determine the dissociation rate constant of kinesins on tau-decorated microtubules, dwell time distributions were fit to a single exponential function according to equation (2):

$$P(t) = A(e^{-kt}) \quad (2)$$

where $P(t)$ is the probability of a kinesin detaching at time t , A is a normalization constant, and k is the dissociation rate constant.

Statistical analysis

All data were presented with error bars indicating either SEM, SD, or 95% CI as specified in the figure legends. All n and number of replicates were mentioned in the figure legends. Sample variance significance was determined using a two-sample F -test for equal variances in MATLAB. The Wilcoxon signed rank test to test the pairwise comparison of tau intensity in different axonal regions was performed in MATLAB. Bootstrapping analysis was performed in MATLAB and used to test for statistical significance and determine confidence intervals.

Data availability

Source data files for all figures and extended data figures, are available with this manuscript.

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Additional information

Author contributions

D. B., C. L. B., and A. G. H. conceptualization; D. B. formal analysis; D. B. investigation; D. B. and A. G. H. methodology; D. B. and A. G. H. project administration; D. B. and A. G. H. software; D. B. validation; D. B. visualization; D. B. and A. G. H. writing—original draft; D. B., C. L. B., and A. G. H. writing—review and editing; C. L. B. and A. G. H. funding acquisition; C. L. B. and A. G. H. resources; A. G. H. supervision.

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Additional files

[Supplementary Material](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

This work by Beaudet and colleagues aims at exploring the effect of phosphorylation on the formation of tau envelopes and consequently on axonal transport, both in vitro on reconstituted microtubules and in human excitatory neurons derived from iPSCs.

The authors found that a relatively widely used construct in which 14 serine or threonine residues, often hyperphosphorylated in Alzheimer's disease, are mutated to alanines (phosphodeficient), increases the density of tau envelopes compared to wildtype tau, whereas a phosphomimetic (same residues mutated to glutamic acid) reduces envelope density both in vitro and in human excitatory neurons derived from iPSCs.

By analysing the trafficking of different kinesins (KIF1a and KIF5C), they observed different effects of tau phosphorylation status on the movement of these two motors.

They then analyse transport of lysosomes by employing live imaging of lysotracker in human excitatory neurons derived from iPSCs transfected with wildtype, phosphodeficient or phosphomimetic tau, observing that phosphodeficient tau seems to reduce transport of lysosomes while phosphomimetic increases transport compared to wildtype tau.

Strengths:

- (1) The work aims to study a novel and underexplored topic in the tau field, tau envelopes, and investigate their relevance to Alzheimer's disease pathology.
- (2) Experiments are well conducted and of high quality.

Weaknesses:

Relying only on in vitro reconstituted microtubules and human neurons derived from iPSCs leaves some doubts about the relevance of these results for Alzheimer's disease, considering the embryonic state of iPSCs-derived neurons.

<https://doi.org/10.7554/eLife.110011.1.sa2>

Reviewer #2 (Public review):

This manuscript examines how disease-associated hyperphosphorylation disrupts tau's role as a cooperative microtubule-binding regulator of intracellular transport. Using in vitro reconstitution assays and live-cell imaging in iPSC-derived neurons, the authors employ phosphomutant tau constructs (E14 to mimic hyperphosphorylation, AP to prevent phosphorylation) at 14 disease-associated residues to isolate phosphorylation effects independent of expression system-dependent PTM heterogeneity. The results show that hyperphosphorylated tau fails to form cooperative envelope-like structures on microtubules, instead binding diffusely and dissociating rapidly. In contrast, wild-type and phospho-resistant tau form cohesive envelopes that regulate motor protein access. At the single-molecule level, hyperphosphorylation reduces KIF5C inhibition while maintaining or enhancing KIF1A inhibition through altered processivity and detachment rates. In live neurons, hyperphosphorylated tau phenocopies tau knockout conditions, weakening tau-mediated inhibition of lysosome transport and increasing processive motility. The authors quantify tau binding using Gaussian mixture model-based image analysis and measure tau kinetics via FRAP, demonstrating that hyperphosphorylation-induced loss of cooperative binding correlates with dysregulated organelle transport. These findings establish a mechanism by which phosphorylation-driven disruption of tau's gatekeeper function on microtubules compromises axonal transport prior to aggregation in tauopathies. The paper provides interesting new knowledge for the field, but there are outstanding concerns that could be further addressed by the authors to strengthen and clarify the current manuscript:

(1) Lack of Phosphatase-Treated Control and Explicit WT Phosphorylation Quantification

Wild-type tau expressed in insect and mammalian cells is known to be phosphorylated by endogenous kinases (eg, GSK3, CDK5, MARK). The manuscript acknowledges this in the Discussion but provides no phosphatase-treated lysate control or quantification of endogenous phosphorylation on WT tau via phospho-specific Western blots. This leaves ambiguity about whether observed differences between WT and E14 reflect purely the introduced mutations or confounding baseline differences in phosphostate content.

(2) Limited Normalization of Motor Effects to Measured Tau Lattice Occupancy

Although kinesin trajectories are classified inside vs. outside tau envelopes (inherently normalizing to local tau density), motor parameters are not systematically reported as functions of tau fluorescence intensity across all constructs. Co-purifying MAPs or microtubule-modifying enzymes in cell lysates is not quantified or excluded, leaving residual uncertainty about tau-specificity of observed motor inhibition. This should be at least acknowledged in the results section.

(3) Insufficient Citation of Prior Neuronal Tau Envelope Evidence

In the Introduction, the authors state, "it was an open question if tau forms envelopes in neurons," but this understates existing evidence. Tan et al. (2019) report tau neuronal staining consistent with envelope formation, while Siahaan et al. (2021) provide more direct evidence in non-neuronal cells. The framing should acknowledge and integrate these prior findings.

(4) Unclear Wording on Expression System-Dependent Phosphorylation

The sentence "The phosphostate of tau is strongly dependent on the expression system" requires rewording. It is ambiguous whether this refers to the final phosphostate achieved after expression or the inherent phosphorylating capacity of each system. Clearer language would strengthen the methodological justification.

(5) Insufficient Quantification of Motor and Lysosome Transport Effect Magnitudes in Results Section

The data on molecular motor motility and lysosome transport are densely described. The magnitude of effects (fold-changes, percentage differences) should be explicitly stated in the Results section when first presenting findings to orient readers to biological significance. For example, effect magnitudes for lysosome run lengths, velocities, and directional bias should be quantified in text, not left to figure inspection.

(6) Incomplete Discussion of Projection Domain Necessity for Envelope Formation

The Discussion states the projection domain is "a critical regulator of both tau-tau and tau-microtubule interactions," but does not engage with prior domain dissection work. Tan et al. (2019) found that the entire projection domain is not necessary for envelope formation in vitro. The authors should discuss which projection domain regions are specifically regulated by phosphorylation vs. required for cooperativity, providing a more nuanced interpretation than implied by their current framing.

<https://doi.org/10.7554/eLife.110011.1.sa1>

Author response:

We thank the reviewers for their thoughtful and constructive feedback. Addressing these points will strengthen the manuscript and improve its clarity.

A primary concern involved the justification for using COS7 cell lysates in reconstitution approaches and iPSC-derived neuronal model systems as models for AD. We will clarify the language throughout the manuscript to more explicitly state the study's goals, emphasize that these systems were selected as robust, well-controlled platforms to test the mechanisms through which tau hyperphosphorylation affects microtubule interactions and tau's role in regulating intracellular transport, and the limitations of in vitro and iPSC models.

Reviewers also raised the possibility that background phosphorylation could contribute to the effects observed in the pseudo-phosphorylation model. We cite two recent preprints that provide insight into this question through quantitatively assessing tau phosphorylation across expression systems. In the revised manuscript, we will elaborate on how their assessment of tau phosphorylation fits within the scope of our approach and clarify how our experimental controls effectively minimize uncertainty related to background phosphorylation.

Another point concerned the potential influence of other microtubule-associated proteins in lysates and the impact of tau lattice occupancy on motility outcomes. To further strengthen this aspect, we will include additional analyses correlating tau intensity along microtubules

with kinesin intensity and motility behavior, and we will more clearly explain how the AP and WT controls provide confidence in the robustness of the system.

Detailed responses to each reviewer comment are provided below point by point. The planned revisions, which include clearer language, stronger justification of the experimental approaches, and additional supporting analyses, will substantially improve the clarity, rationale, and overall impact of the study.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This work by Beaudet and colleagues aims at exploring the effect of phosphorylation on the formation of tau envelopes and consequently on axonal transport, both in vitro on reconstituted microtubules and in human excitatory neurons derived from iPSCs.

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Weaknesses:

Relying only on in vitro reconstituted microtubules and human neurons derived from iPSCs leaves some doubts about the relevance of these results for Alzheimer's disease, considering the embryonic state of iPSCs-derived neurons.

We agree with the reviewer that iPSC-derived neurons represent an immature state compared with the neurons affected in Alzheimer's disease. However, iPSC-derived neurons, together with in vitro reconstitution, provide insight into (1) whether tau hyperphosphorylation influences its association with microtubules and its ability to form envelope-like structures thought to regulate transport, (2) how tau hyperphosphorylation affects the motility of kinesin motors that are strongly inhibited by tau, and (3) how transport of endogenous degradative organelles such as lysosomes are impacted by tau hyperphosphorylation. We hope that our studies will help to inform future studies examining how tau-related dysfunction evolves in more mature neurons and contributes to the more severe pathological effects observed at later disease stages.

We will include a paragraph in the Discussion section addressing the limitations of this study to better contextualize our findings within the broader effort to understand tauopathies and

Alzheimer's disease.

Reviewer #2 (Public review):

This manuscript examines how disease-associated hyperphosphorylation disrupts tau's role as a cooperative microtubule-binding regulator of intracellular transport. Using in vitro reconstitution assays and live-cell imaging in iPSC-derived neurons, the authors employ phosphomutant tau constructs (E14 to mimic hyperphosphorylation, AP to prevent phosphorylation) at 14 disease-associated residues to isolate phosphorylation effects independent of expression system-dependent PTM heterogeneity. The results show that hyperphosphorylated tau fails to form cooperative envelope-like structures on microtubules, instead binding diffusely and dissociating rapidly. In contrast, wild-type and phospho-resistant tau form cohesive envelopes that regulate motor protein access. At the single-molecule level, hyperphosphorylation reduces KIF5C inhibition while maintaining or enhancing KIF1A inhibition through altered processivity and detachment rates. In live neurons, hyperphosphorylated tau phenocopies tau knockout conditions, weakening tau-mediated inhibition of lysosome transport and increasing processive motility. The authors quantify tau binding using Gaussian mixture model-based image analysis and measure tau kinetics via FRAP, demonstrating that hyperphosphorylation-induced loss of cooperative binding correlates with dysregulated organelle transport. These findings establish a mechanism by which phosphorylation-driven disruption of tau's gatekeeper function on microtubules compromises axonal transport prior to aggregation in tauopathies. The paper provides interesting new knowledge for the field, but there are outstanding concerns that could be further addressed by the authors to strengthen and clarify the current manuscript:

(1) Lack of Phosphatase-Treated Control and Explicit WT Phosphorylation Quantification

Wild-type tau expressed in insect and mammalian cells is known to be phosphorylated by endogenous kinases (eg, GSK3, CDK5, MARK). The manuscript acknowledges this in the Discussion but provides no phosphatase-treated lysate control or quantification of endogenous phosphorylation on WT tau via phospho-specific Western blots. This leaves ambiguity about whether observed differences between WT and E14 reflect purely the introduced mutations or confounding baseline differences in phosphostate content.

Tau contains ~85 putative phosphorylation sites and is modified by several kinases in cells. Studies by Siahaan et al. (2024) and Fan et al. (2025) provide detailed insight into tau phosphorylation, its role in protecting the microtubule lattice from severing enzymes, and the implications of phosphorylation patterns for aggregate formation. Specifically, Fan et al. (2025) show that HEK-expressed tau is phosphorylated by endogenous kinases at 58 residues, with most phospho-occupancy levels below 15%, indicating substantial heterogeneity among individual tau molecules. In the revised manuscript, we will (1) provide justification for the use of the pseudo-phosphorylation model system as an approach to limit heterogeneity among tau molecules, (2) clarify the importance of the WT and AP controls, (3) discuss that E14, WT, and AP tau likely exhibit similar degrees of background phospho-heterogeneity, with WT tau likely exhibiting some overlap between background phosphorylation and the 14 AD-associated sites examined, and (4) expand the discussion to emphasize that although background phosphorylation is present, our results do not suggest that it contributes significantly to the observations reported in this study.

(2) Limited Normalization of Motor Effects to Measured Tau Lattice Occupancy

Although kinesin trajectories are classified inside vs. outside tau envelopes (inherently normalizing to local tau density), motor parameters are not systematically reported as functions of tau fluorescence intensity across all constructs. Co-purifying MAPs or microtubule-modifying enzymes in cell lysates is not quantified or excluded, leaving

residual uncertainty about tau-specificity of observed motor inhibition. This should be at least acknowledged in the results section.

The reviewer raises a valid point. It is challenging to compare conditions where the occupancy of tau on microtubules is similar across conditions, as phosphorylation strongly effects the interaction between tau and microtubules. We will quantify and report tau intensity in single-molecule motility assays. On the second point, while effects from other MAPs or motor proteins could potentially affect kinesin motility, we would expect that these effects would be similar for all tau phosphomutant constructs, such that the effect of tau phospho-states on kinesin motility can be assessed.

(3) Insufficient Citation of Prior Neuronal Tau Envelope Evidence

In the Introduction, the authors state, "it was an open question if tau forms envelopes in neurons," but this understates existing evidence. Tan et al. (2019) report tau neuronal staining consistent with envelope formation, while Siahaan et al. (2021) provide more direct evidence in non-neuronal cells. The framing should acknowledge and integrate these prior findings.

We agree with the reviewer that evidence from several studies using reconstitution systems, fixed neurons, and live cultured cells provides evidence of tau envelope formation in neurons. Specifically, tau envelopes have been observed along taxol-stabilized or GMPCPP-capped GDP microtubules in vitro (e.g., Dixit et al., 2008; Monroy et al., 2018; Tan et al., 2019; Siahaan et al., 2019), in 4% PFA-fixed and Triton X-100-extracted DIV7 mouse hippocampal neurons (Tan et al., 2019), and in live, non-neuronal U-2 OS cells following taxol treatment (Siahaan et al., 2022) or elevated pH (Siahaan et al., 2024). However, to our knowledge, our study is the first to demonstrate tau envelope formation in live neuronal cells under normal cell culture conditions. We will revise this sentence in the manuscript to more precisely position our findings within the context of prior studies.

(4) Unclear Wording on Expression System-Dependent Phosphorylation

The sentence "The phosphostate of tau is strongly dependent on the expression system" requires rewording. It is ambiguous whether this refers to the final phosphostate achieved after expression or the inherent phosphorylating capacity of each system. Clearer language would strengthen the methodological justification.

We agree that the wording here is ambiguous and requires clarification. In the revised manuscript, we will clarify that tau phosphorylation depends on the expression system used; bacterial systems lack the capacity for many post-translational modifications compared with insect and mammalian systems. We will also emphasize that in insect and mammalian expression systems, tau phosphorylation occurs heterogeneously, as demonstrated in previous studies by Siahaan et al. (2024) and Fan et al. (2025).

(5) Insufficient Quantification of Motor and Lysosome Transport Effect Magnitudes in Results Section

The data on molecular motor motility and lysosome transport are densely described. The magnitude of effects (fold-changes, percentage differences) should be explicitly stated in the Results section when first presenting findings to orient readers to biological significance. For example, effect magnitudes for lysosome run lengths, velocities, and directional bias should be quantified in text, not left to figure inspection.

Our initial justification for omitting quantitative data from the results text was to improve readability; however, in doing so, we may have reduced the accessibility and clarity regarding the significance of the findings. In the revised manuscript, we will incorporate the relevant quantifications and statistical significance for the motility data in the text.

(6) Incomplete Discussion of Projection Domain Necessity for Envelope Formation

The Discussion states the projection domain is "a critical regulator of both tau-tau and tau-microtubule interactions," but does not engage with prior domain dissection work. Tan et al. (2019) found that the entire projection domain is not necessary for envelope formation in vitro. The authors should discuss which projection domain regions are specifically regulated by phosphorylation vs. required for cooperativity, providing a more nuanced interpretation than implied by their current framing.

We agree with the reviewer. Tan et al. (2019) demonstrated that the proline-rich region (residues 198–244) within the projection domain of full-length 2N4R tau is the minimal region required to maintain tau's ability to form envelopes along microtubules. We will incorporate this work on the dissection of the projection domain and discuss how the phosphorylation sites examined in our study are primarily located within this region. Together, these data highlight the proline-rich region as a potential major regulator of tau–tau cooperativity.

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